CS 350: Operating Systems
Fall '22
Lecture: MWF 1:10-2:10pm, CW 212
Lab: Th 11:40-1:05, EB N01
Instructor: Jayson Boubin

Contact Information:
Jayson Boubin
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Graduate TA:
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Description:

Introduction to operating system concepts: processes, CPU scheduling, memory management, file system and storage, concurrency, and multi-threaded programming.

Prerequisites:

- CS 220 Computer Organization and Assembly Language
- Familiarity with the C programming language and the ability to work in a Linux environment.

Course Goals:

- Understanding of process concepts and CPU scheduling.
- Understanding of the memory hierarchy and memory management.
- Familiarity with process control blocks, system calls, context switching, interrupts, and exception control flows.
- Familiarity with process synchronization, inter-process communication, and threads.
- Familiarity with multi-threaded programming.
- Familiarity with file systems and disk scheduling algorithms and I/O.
- Understanding of basic security concepts.

Text:

This course requires a **free** textbook. You do not need to purchase any textbooks for this course.

- Required: Operating Systems: Three Easy Pieces
  - Note: Available for free online and on the course website
- Recommended: Modern Operating Systems, Andrew Tanenbaum, Prentice Hall; 4th edition.

Topics:

- Introduction to operating systems
- Process concepts, scheduling, synchronization, deadlocks
- Multi-thread programming
- Memory hierarchy and memory management
- File systems
- Disk allocation, scheduling
- I/O
- Security

Grading Policy:

Lab Assignments	20%
Homework	25%
Midterm 1	10%
Midterm 2	15%
Final Exam	25%
Attendance	5%

Notes:

- Homework: Homework will be assigned in class and is due at ***11:59pm*** on the due date specified on the assignment. Late homework can then be submitted up to 48 hours after the specified due date for a ***30%*** penalty of the point value of the assignment. No assignments will be accepted after that time. Homework must be the student's own work. Homework must be typed except for needed diagrams. There will be 3 - 4 homework assignments. Graded homework assignments will be reviewed in class or lab periods.
- Labs: Programming assignments/labs will be assigned during lab periods. Labs must be submitted online by ***11:59pm*** on the date due as specified on the assignment. Labs are subject to the same late penalties as homework. There will be a total of 4 labs.
- Midterms: The midterms are closed notes, closed book. No calculators are permitted. Midterms will be taken in class
- Final Exam: The final exam is comprehensive and is closed notes, closed book. No calculators are permitted. The exam date and time

is determined by the university. Final exams may not be taken early.

- ***Note: If there are unforeseen emergencies that impact the grading scheme, appropriate adjustments will be made. An example of an unforeseen emergency would be a fire alarm going off during the final exam.***

Attendance:

Attendance will be taken at the start of class. It is the student's responsibility to find out what material was missed or assignments given if unable to attend class. Makeup exams require that appropriate documentation be provided ahead of time and are up to the discretion of the instructor. Attendance accounts for 5 percent of the course grade. Each student is afforded 2 unexcused absences from lecture or lab periods. For each additional absences, 1% will be deduced from the students final course grade.

Academic Misconduct:

Students are required to follow the Binghamton University "Code of Student Conduct" and the Watson College "Student Academic Honesty Code" which can be found at the following links:

https://www.binghamton.edu/student-conduct/binghamton-student-code.pdf

https://www.binghamton.edu/watson/about/academic-honesty.html

Among the restrictions, students are prohibited from:

- Providing or receiving information during exams
- Receiving assistance on homework and lab problems. Providing assistance on homework and lab problems to students in the course during this or any other semester.
- Submitting plagiarized (i.e. copied but unacknowledged) work for credit

Depending on the violation, misconduct will result in grade reductions for the assignments in question and or referral to the Watson College Academic Honor Committee.

Communication:

Brightspace will be used to post materials for the course. CS 350 is, however, not an online course.

Brightspace will be used both to post course material and as a student discussion platform for the course. Some examples of acceptable topics to discuss include; general information, concepts as related to lecture or assignments, interpretation of assignments, problems with coding such as syntax and execution errors, etc. ***Please do not post answers to homework problems and do not post any code for the programming assignments.*** *Brightspace* will be monitored. ***Students are responsible for any announcements/info provided on Brightspace.***

Important Dates:

- Monday, September 5th — Labor day, no class
- Tuesday, September 6th — Add/Drop Deadline
- Friday, September 23rd — **Midterm I (Tentative)**
- Monday, September 26th — Rosh Hashanah, no class
- Wednesday, September 28th — Academic Assessment Period
- Wednesday, October 5th — Yom Kippur, no class
- Th-F, October 20-21 — Fall break, no class
- Friday, October 28th — Withdrawal Deadline
- Friday, November 11th — **Midterm II (Tentative)**
- W-F, November 23-25 — Thanksgiving, no class
- Friday, December 9th — Last day of classes
- **Final: TBD**

See Homework and Lab assignments for other due dates.